Native timeout diagnostic: original 139, patched 0

Two fresh bounded full-tool controls reproduce a timeout formatting defect in pinned YAFU c7d90283972820f99b3b4920bfc1eff05c1f6342. The original exits 139; the patched executable exits 0 and logs its integer timeout. Both use the exact R05 444-bit residual and selected polynomial. This diagnoses the failure mechanism without claiming a production backtrace or a new factor.

Fixed input and Q range; one bounded native control per binary

Original timeout path

uint32_t passed to string conversion

Native executable: exit 139

Extracted statement: exit 139

Corrected timeout path

unsigned integer conversion

Native executable: exit 0

Integer timeout logged; no new factor

Run 2026-10-23 · 1 s native limit · 30 s outer bound · no production backtrace

Recorded controls

Control	Observed result
Original sieving printf, format-as-error compile	rejected: uint32_t passed to string conversion
Original second timeout printf, format-as-error compile	rejected: uint32_t passed to floating conversion
Original extracted sieving statement, one run	exit 139
Both corrected extracted statements	compile and run successfully
Original full native NFS, one bounded control	exit 139
Patched full native NFS, one bounded control	exit 0; integer timeout message logged
Checked resume, prepare only	installed validated binary; native arithmetic replay passes
Wrong binary or active native lock	both rejected

The raw native summary binds the actual executable hashes. The original matches the production R05 binary. The patch changes two timeout messages only: the sieving message's %1.4sec parses as a string conversion %1.4s, and the other message's %1.4f expects a double. Both incorrectly receive a uint32_t. Corrected messages use %u with an unsigned cast. Historical build warnings already identify both defects.

Inputs, bounds and evidence

The protocol limits extracted controls to five seconds each with core dumps disabled. The separate full-tool protocol allows two fresh native controls, two threads, seed 20261022, native limit one second, Q range 2935000 to 2935500, sieving phase only, outer bound 30 seconds plus five-second kill grace, 4 GiB address space and 64 MiB per output file. The compatible local host has four CPUs and 16 GiB RAM, L0. There is one control per native arm, no replicate interval and no timing comparison; the diagram shows recorded outcomes, not performance.

The original R05 relations and checkpoint were never imported or modified. Each control uses its own selected job, parameters, output and native executable. Both complete within the outer bound. The patched tool's final C134 label repeats the unchanged input and is not a proper factor. The first separate build attempt fails to autodetect GMP's multiarch headers; the second explicitly sets the include path and succeeds. Raw build errors, commands, compiler versions, sources/hashes, diagnostics, output/state and wrapper controls are retained in the sibling raw archive.

The checked wrapper installs only a binary matching a passing control summary, under both existing locks, with the previous executable retained. Its copied executable is rehashed before installation. Prepare-only replay verifies the unchanged original target and prime certificates. The R04 compact checkpoint is used only for this non-search preparation control; the actual later card binds the newer R05 relation checkpoint after required artifact-store publication.

Scope and remaining work

The formatting defect and its corrected bounded native behavior are verified. The exact R05 production instruction pointer remains unobserved. No longer production allocation was run. Eight of nine integers remain fully factored, with the 444-bit residual still composite. This diagnostic adds no algebraic identity or Lean theorem; existing partition proofs and native prime certificates remain applicable. Existing mathematical and performance graphs have no changed result and remain unchanged.

The backlog card now requires the checked tool and the R05 relation archive, but remains draft and unqueued pending actual store publication, a merged source pin and selected lab. No paid resource, task or lease was created. The next production run must preserve all relation state and independently certify any proper factor it returns.